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1 Question 1

Fig. 1 shows the normals computed in CloudCompare for different (increasing) neighborhood radii. We take Fig. 1 (b) as a baseline visualization and highlight the effect of radius size in what follows:

- Too small radius: The normal is estimated using a very small neighborhood that usually contains an insufficient number of points. Consequently, the estimation becomes sensitive to noise and outliers, thereby resulting in irregular and unstable normals. This behavior is clearly observed in Fig. 1 (a). Specifically, the facade, which should appear uniformly colored (just like in other visualizations), is instead colored with a mixture of red and yellow Dip values, indicating inconsistent normal orientations. Moreover, parts of the ground surface appear green, even though the ground is globally blue. These green regions correspond to sparser areas of the cloud, where the small radius further amplifies variations and causes changes in normals.
- Too big radius: Conversely, in this case, the neighborhood contains a large number of points and may include those belonging to different surfaces. This can further lead to averaging of different structures, which eventually results in smoothing and loss of local information. This effect is visible in Fig. 1 (c) and (d). Compared to the baseline, the window edges appear less distinct and their normals become similar to those of the surrounding facade. This occurs because, within a large neighborhood, the surrounding facade points dominate the estimation and prevail over the window structure. Moreover, we observe "transition" regions between the ground and the facade, as well as between the facade and the roof. These regions appear because the neighborhood contains points from surfaces with different orientations, causing their normals to be averaged into the intermediate Dip value (i.e., blue for horizontal surfaces and red for vertical surfaces, blending into green for inclined orientations). As the radius increases, these regions become wider, since the larger neighborhood captures a broader context and further smooths the orientation difference.

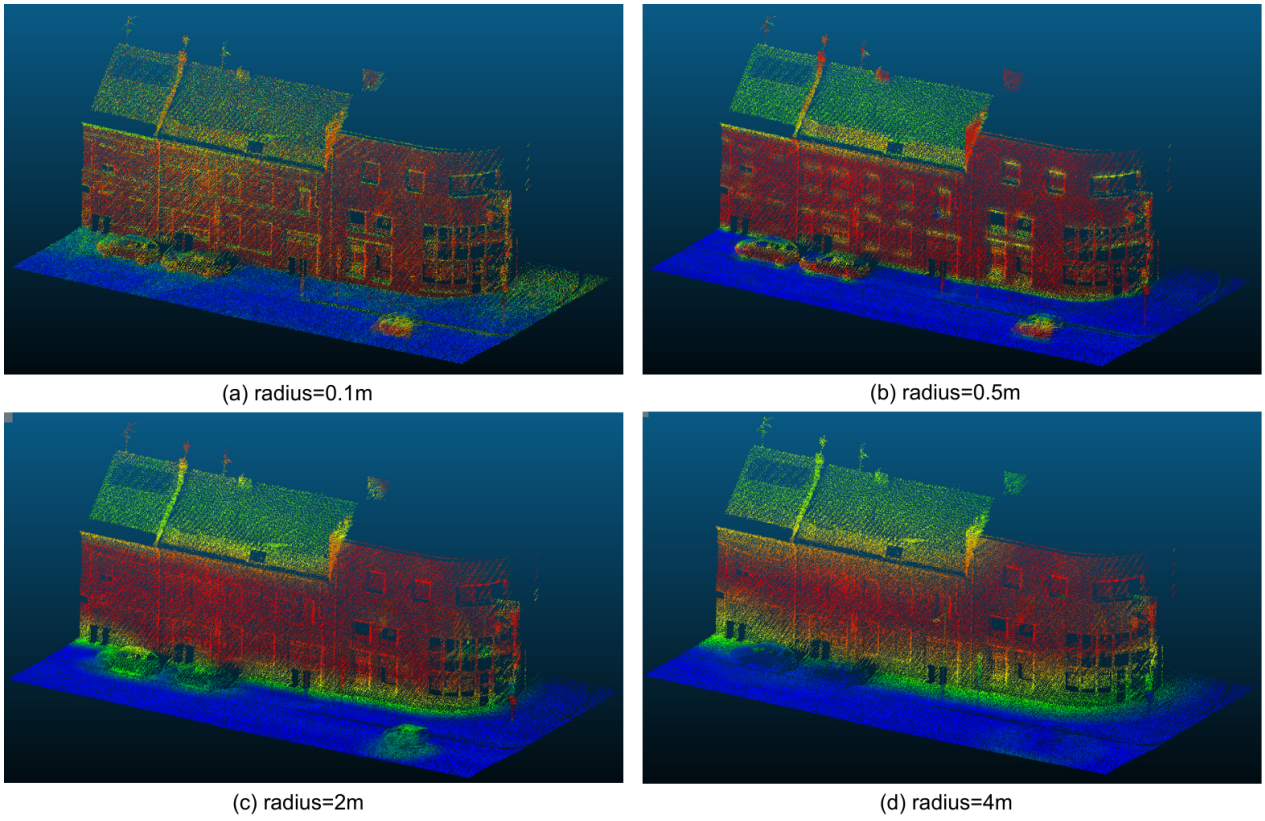


Figure 1: Dip visualization of normals estimated using "Normals" tool in CloudCompare with too small (a), medium/baseline (b), and too big (c, d) neighborhood radii.

2 Question 2

I would say that choosing an appropriate neighborhood scale for normal estimation depends on the characteristics of the point cloud and therefore does not admit a universal solution. In general, as discussed in the previous question, using a smaller neighborhood can introduce noisy normal estimates, while a larger one can oversmooth the local geometry and diminish the details. Therefore, in order to avoid both negative effects, the neighborhood should be big enough to reduce the influence of noise, but at the same time smaller than the structures that we want to preserve. Another factor to consider would be the point density such that when having denser point clouds, we are allowed to use smaller neighborhoods, as they already contain a sufficient number of points, whereas with sparser clouds, we would require larger neighborhoods in order to include enough points. This was also visualized and discussed in the previous question for ground surface in Fig. 1 (a). Given these considerations, I would suggest empirically testing several neighborhood scales and visually assessing the noise presence, detail preservation, and overall quality for a good normal estimation.

If we wanted to avoid empirical tests and use a more quantitative criterion, I would say that we can infer the optimal neighborhood scale from PCA eigenvalues λ_1, λ_2, and λ_3. Since normal estimation is equivalent to finding the best approximated plane in a point neighborhood, an optimal neighborhood scale should yield a locally planar structure, with large values of λ_1 and λ_2, and a small value of λ_3. Since we know that λ_3 corresponds to the least squares plane fitting error and can be interpreted as the distance of points to the approximating plane, a good neighborhood scale should minimize λ_3, while keeping a planarity feature value, i.e. $\frac{\lambda_2-\lambda_3}{\lambda_1}$, high. We will now analyze the values of λ_3 and planarity across different neighborhood radii shown in Fig. 2. As seen in Fig. 2 (a), the mean value of λ_3 increases with the neighborhood radius and becomes significantly larger for a radius of 2.0 m, which indicates deviation from the planarity and suggests that this scale is not optimal. This higher λ_3 value is a consequence of ”transition” regions that we discussed in the previous question. Smaller radii, such as 0.1 m and 0.5 m, yield lower λ_3 values and therefore better plane fitting. However, Fig. 2 (b) shows that the mean planarity is very low for a radius of 0.1 m, which suggests that noise dominates at this scale and prevents good planar estimation. In contrast, a radius of 0.5 m results in the highest planarity value, while larger radii again reduce the planarity. These observations confirm the visual conclusions from the previous question where radius of 0.5 m provided the best compromise between noise reduction and preservation of local structures. Based on this analysis, I would say that the same approach can conversely be applied to other point clouds as well.

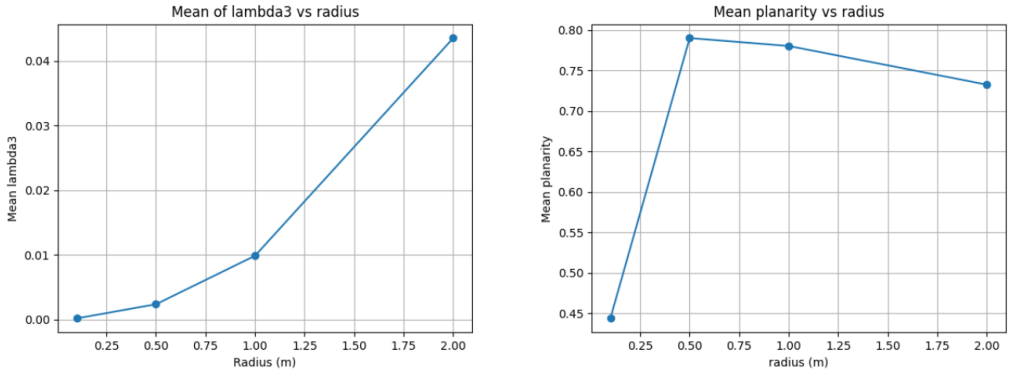


Figure 2: Visualization of (a) the mean value of the PCA eigenvalue λ_3 and (b) the mean planarity for different neighborhood radii (0.1 m, 0.5 m, 1.0 m, and 2.0 m).

3 Question 3

Fig. 3 shows the normals computed using the *compute_local_PCA* function with a spherical neighborhood search where radius=0.5 m. This visualization is equivalent to Fig. 1(b), where normals were estimated using CloudCompare, which therefore confirms the correctness of our implementation.

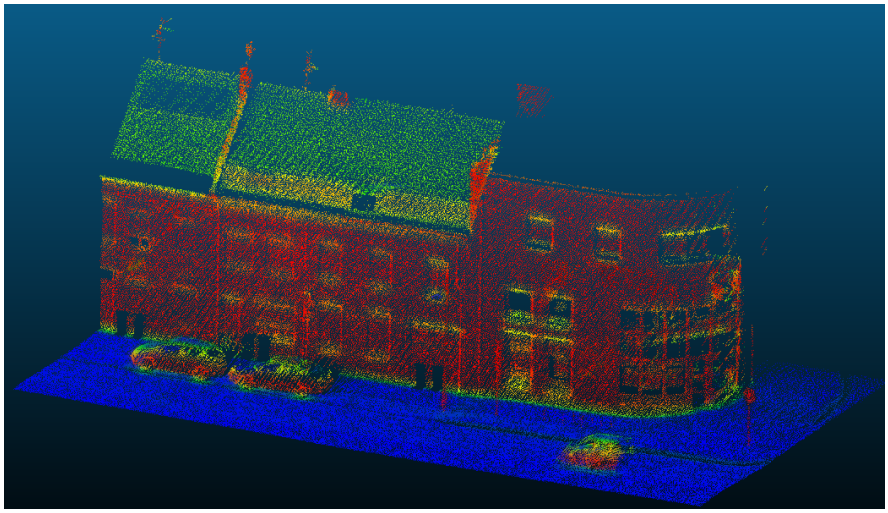


Figure 3: Dip visualization of normals estimated using the PCA function with a spherical neighborhood (radius=0.5 m).

4 Question 4

Fig. 4 shows the normals computed using the *compute_local_PCA* function with a k-nearest neighbors (kNN) search where $k = 30$. Compared to the radius method in Fig. 3, this approach yields globally similar results, with a few local changes detailed below:

- Window and facade area: The regions around windows and facades are part of the high-density areas of the point cloud. In these regions, the kNN approach selects only very local neighbors when computing normals, which then better captures fine details. As a result, window edges are more clearly differentiated from their interior and from the surrounding facade. In contrast, the radius method includes a larger number of points (in these dense regions) and therefore aggregates points from both window frames and facade surfaces. This leads to stronger averaging and smoother normals of this area, thereby reducing the visibility of window details.

- Ground-wall junction: In this area, both methods produce intermediate (green) Dip values due to the presence of two intersecting surface orientations (the ground being horizontal and the wall being vertical). However, this region is slightly wider for the radius method. This is because spherical neighborhoods include all points within a fixed radius, and therefore gather points from both surfaces near the junction independently of their orientation. As a consequence, the normal estimation averages the two surface orientations. On the contrary, the kNN method favors points belonging to the same surface as the query point, since these points are usually closer in Euclidean distance. This leads to neighborhoods dominated by a single surface, which inherently better distinguishes the ground and the wall.

- The curbside: This area corresponds to the region between the road and the sidewalk and differs in color between the two methods. With the radius method, the normal estimation is dominated by a larger number of points from the flat road and sidewalk surfaces, which results in light blue Dip values that are consistent with the surrounding ground. In contrast, the kNN method relies on a relatively small number of neighbors, which leads to a more balanced contribution of points from both this region and the surrounding surface. This produces intermediate Dip values, thereby indicating the presence of a small slope.

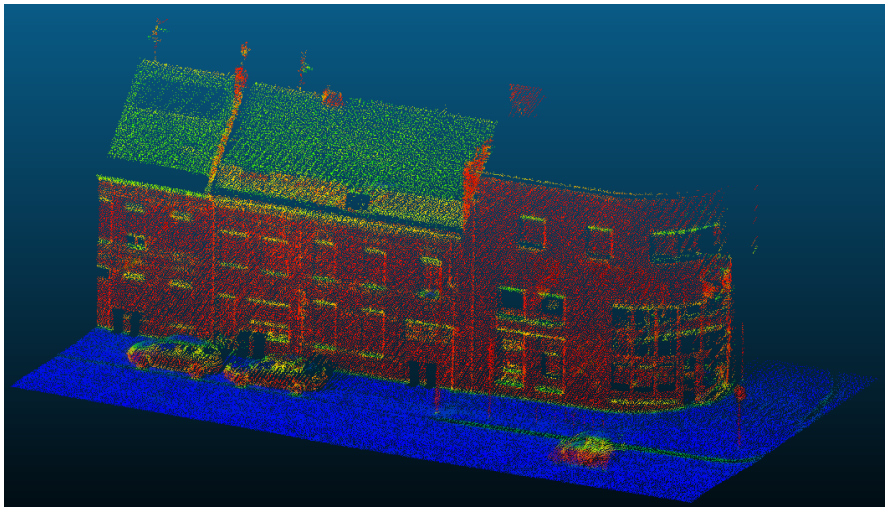
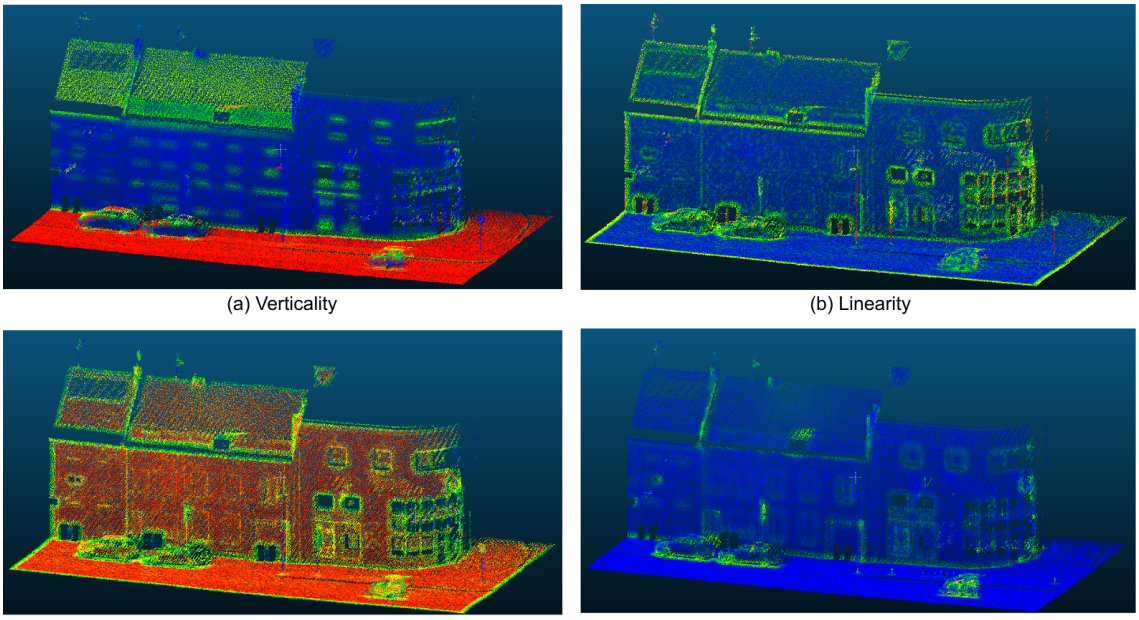


Figure 4: Dip visualization of normals estimated using the PCA function with a k-nearest neighbors search (k=30).

5 Question 5

Fig. 5 shows four geometric features as scalar fields of the point cloud. As mentioned in the assignment, eigenvalues measure the spread of points along the directions given by the eigenvectors. Having that, we now explain the following:

- Linearity: The name comes from a line-like (1D) neighborhood, in which the spread of points is mainly along one direction. In this case, the largest eigenvalue, λ_1, is high, while λ_2 and λ_3 are low. Therefore, $1 - \frac{\lambda_2}{\lambda_1} \approx 1 - 0 \approx 1$ meaning that linearity is high. Conversely, when the neighborhood is planar (2D), λ_2 increases and $\lambda_2/\lambda_1 > 0$, which reduces the linearity value.
- Planarity: The name comes from a planar-like (2D) neighborhood, in which the spread of points is high in two directions. In this case, λ_1 and λ_2 are high, while λ_3 is low. Therefore, $\frac{\lambda_2 - \lambda_3}{\lambda_1} \approx \frac{\lambda_2}{\lambda_1} \approx 1$ meaning that planarity is high. Again, conversely, for 1D neighborhoods where λ_2 and λ_3 are small, the planarity value decreases.
- Sphericity: The name comes from a spherical-like (3D) neighborhood, where the spread of points is similar and high in all directions ($\lambda_1 \approx \lambda_2 \approx \lambda_3$). Therefore, $\frac{\lambda_3}{\lambda_1} \approx 1$, meaning that the sphericity value is high. Conversely, for 1D or 2D neighborhoods, λ_3 is small, which reduces the sphericity value.



(a) Verticality (b) Linearity

(c) Planarity (d) Sphericity

Figure 5: Visualization of geometric features.